

Formalisms Every Computer Scientist Should Know: Homework Solutions

1 Homework 1

Theorem 1.1. *If there is a one-to-one function on a set A to a subset of a set B and there is also a one-to-one function on B to a subset of A , then A and B are equipollent.*

Proof. We need to show that there exists a bijection between A and B .

Assume such two functions exist. Fix the two functions $\hat{f} : A \rightarrow B$ and $\hat{g} : B \rightarrow A$ be one-to-one functions.

Then there exist bijections $\hat{f}^{-1} : \hat{f}(A) \rightarrow A$ and $\hat{g}^{-1} : \hat{g}(B) \rightarrow B$.

Define $A_C \subseteq A$ as the set of such elements $a \in A$ that $(\hat{g} \circ \hat{f})^n a = a$ for some integer $n \in \mathbb{N}$ (zero is not in $\mathbb{N}$).

In the same way, define $B_C = \{b \in B \mid \exists n \in \mathbb{N} : (\hat{f} \circ \hat{g})^n b = b\}$.

For a given element $a \in A$, define a sequence of ancestors $X(a) = (a, \hat{g}^{-1}(a), \hat{f}^{-1}(\hat{g}^{-1}(a)), \dots)$.

The sequence is finite if the last element is in $A \setminus \hat{g}(B)$ or in $B \setminus \hat{f}(A)$, and infinite otherwise. In the same way, for $b \in B$, define $X(b) = (b, \hat{f}^{-1}(b), \hat{g}^{-1}(\hat{f}^{-1}(b)), \dots)$.

Define $A_I = \{a \in A \mid X(a) \text{ is infinite}\}$, $A_E = \{a \in A \mid |X(a)| \text{ is even}\}$, $A_O = \{a \in A \mid |X(a)| \text{ is odd}\}$. Observe that A_I, A_E, A_O are disjoint and cover A . Similarly, define B_I, B_E, B_O .

Notice that $A_C \subseteq A_I$, since for any $a \in A_C$, $X(a)$ is periodic and infinite.

I claim that $\hat{f}$ maps A_I bijectively onto B_I . To see this, we need to show that for any $b \in B_I$, there exists a preimage $a \in A_I$ such that $\hat{f}(a) = b$. Such a is the second element of $X(b)$, $a = \hat{f}^{-1}(b)$. Then $\hat{f}(a) = b$. a is indeed in A_I , because $X(a) = (a = \hat{f}^{-1}(b), \hat{g}^{-1}(\hat{f}^{-1}(b)), \dots)$ is also infinite.

Next, show that $\hat{f}$ maps A_E bijectively onto B_O . To see this, we need to show that for any $b \in B_O$, there exists a preimage $a \in A_E$ such that $\hat{f}(a) = b$. Such a is the second element of $X(b)$, $a = \hat{f}^{-1}(b)$. Then $\hat{f}(a) = b$. a is indeed in A_E , because $X(a) = (a = \hat{f}^{-1}(b), \hat{g}^{-1}(\hat{f}^{-1}(b)), \dots)$. Then $|X(a)| = |X(b)| - 1$ is even.

In the same way, $\hat{g}$ maps B_E bijectively onto A_O . Therefore, $\hat{g}^{-1}$ maps A_O bijectively onto B_E .

Define a map $h : A \rightarrow B$:

$$h(a) = \begin{cases} \widehat{f}(a), & a \in A_I \\ \widehat{f}(a), & a \in A_E, \\ \widehat{g}^{-1}(a), & a \in A_O \end{cases} \quad (1)$$

which is the desired bijection between A and B . $\square$

2 Homework 2

Theorem 2.1 (Knaster-Tarski). *Let $(A, \sqsubseteq)$ be a complete lattice and let F be a monotone function on A . Further, let $\hat{y} = \bigsqcup \{x \mid x \sqsubseteq F(x)\}$ and $\hat{z} = \bigsqcap \{x \mid F(x) \sqsubseteq x\}$. It holds, that:*

1. $\hat{y}$ and $\hat{z}$ are fixpoints of F ,
2. for all fixpoints x of F , $\hat{z} \sqsubseteq x \sqsubseteq \hat{y}$

Proof. Suppose $(A, \sqsubseteq)$ is a complete lattice and let F be a monotone function on A . Let U be the set of elements $x \in A$ for which $x \sqsubseteq F(x)$ and let D be the set of elements $x \in A$ for which $x \sqsupseteq F(x)$. Then as described in the lecture let us denote the join and meet of these two sets respectively as $\hat{y} = \bigsqcup U$ and $\hat{z} = \bigsqcap D$ (this meet and join exist because the considered lattice is complete). We shall prove:

1. $\hat{y}$ and $\hat{z}$ are fixpoints of F .
2. For all fixpoints x of F the following holds $\hat{z} \sqsubseteq x \sqsubseteq \hat{y}$.

1. Let us first prove that $\hat{y}$ is a fixpoint of F . Let us pick an arbitrary element $u \in U$, then by definition of $\hat{y}$ we have $u \sqsubseteq \hat{y}$. Because F is monotone this implies $F(u) \sqsubseteq F(\hat{y})$. And because u was picked arbitrarily from U this means that $F(\hat{y})$ is an upper bound on U . But $\hat{y}$ is the least upper bound of U and thus $\hat{y} \sqsubseteq F(\hat{y})$. Again by applying the monotonicity of F we get $F(\hat{y}) \sqsubseteq F(F(\hat{y}))$ and we can conclude $F(\hat{y}) \in U$. But because $\hat{y}$ is the join of U and $F(\hat{y})$ is an element of U it follows $F(\hat{y}) \sqsubseteq \hat{y}$. Together with the inequality $\hat{y} \sqsubseteq F(\hat{y})$ this gives us $F(\hat{y}) = \hat{y}$, hence $\hat{y}$ is a fixpoint of F . The proof that $\hat{z}$ is a fixpoint of F can be done along the same lines by considering $\sqsupseteq$ instead of $\sqsubseteq$ and the set D instead of U .

2. Suppose $\hat{x}$ is a fixpoint of F . Then clearly $\hat{x} \in U$ and hence $\hat{x} \sqsubseteq \hat{y}$ as $\hat{y}$ is the join of U . Similarly $\hat{x} \in D$ and hence $\hat{x} \sqsupseteq \hat{z}$ as $\hat{z}$ is the meet of D . So it follows that $\hat{z} \sqsubseteq \hat{x} \sqsubseteq \hat{y}$ and the proof is done. $\square$

3 Homework 4

3.1 Sorting

Definition 3.1 (ordered). A function $\text{ordered} : \Sigma^* \rightarrow \{\text{true}, \text{false}\}$ is defined as follows. $\text{ordered}(\varepsilon) = \text{true}$, $\text{ordered}(a) = \text{true}$, $\text{ordered}(abx) = (a \leq b) \wedge \text{sorted}(bx)$, where $a \in \Sigma$, $x \in \Sigma^*$.

Lemma 3.1. $\text{sorted}(\text{ms}(x)) = \text{true}$.

Proof. We will use induction on the length of x . Obviously, $\text{sorted}(\text{ms}(\varepsilon)) = \text{sorted}(\text{ms}(a)) = \text{true}$.

Recall that $\text{ms}(x) = \text{merge}(\text{ms}(\text{odd}(x)), \text{ms}(\text{even}(x)))$. By the induction assumption, we know that $\text{ms}(\text{odd}(x)), \text{ms}(\text{even}(x))$ are sorted, since their length is less than the length of x . Now it is enough to show that $\text{sorted}(\text{merge}(y, z)) = \text{true}$ if y and z are sorted. Consider $\text{sorted}(\text{merge}(ay, bz))$. Without loss of generality, let $a \leq b$, so $\text{sorted}(\text{merge}(ay, bz)) = \text{sorted}(\text{amerge}(y, bz)) = \text{true}$, since a is less or equal to both b and the next element of y , and $\text{sorted}(\text{merge}(y, bz)) = \text{true}$ holds by induction. $\square$

Definition 3.2 (permutation). For all natural n , the function $\text{permutation} : \Sigma^n \times \Sigma^n \rightarrow \{\text{true}, \text{false}\}$ is defined as follows. $\text{permutation}(\varepsilon, \varepsilon) = \text{true}$, $\text{permutation}(a, b) = (a == b)$, $\text{permutation}(ax, yaz) = \text{permutation}(x, yz)$, and if $a \notin y$ then $\text{permutation}(ax, y) = \text{false}$, where $a, b \in \Sigma$, $x, y, z \in \Sigma^*$.

Lemma 3.2. $\text{permutation}(x, \text{ms}(x)) = \text{true}$.

Proof. We will use induction on the length of x . Obviously, $\text{permutation}(\varepsilon, \text{ms}(\varepsilon)) = \text{permutation}(a, \text{ms}(a)) = \text{true}$.

First, let's show that $\text{merge}(x, y)$ is a permutation of xy . Indeed, if, for instance, $a \leq b$, then $\text{permutation}(\text{merge}(ax, by), axby) = \text{permutation}(\text{amerge}(x, by), axby) = \text{permutation}(\text{merge}(x, by), xby) = \text{true}$ by induction.

Finally, $\text{ms}(x) = \text{merge}(\text{ms}(\text{odd}(x)), \text{ms}(\text{even}(x)))$. By the induction assumption, we know that $\text{ms}(\text{odd}(x)), \text{ms}(\text{even}(x))$ are permutations of $\text{odd}(x), \text{even}(x)$ respectively, since their length is less than the length of x . Moreover, as we have already shown above, $\text{merge}(\text{ms}(\text{odd}(x)), \text{ms}(\text{even}(x)))$ is a permutation of $\text{ms}(\text{odd}(x))\text{ms}(\text{even}(x))$, which is a permutation of $\text{odd}(x)\text{even}(x)$. Finally, it is sufficient to prove that $\text{odd}(x)\text{even}(x)$ is a permutation of x . Using induction, $\text{permutation}(\text{odd}(ax)\text{even}(ax), ax) = \text{permutation}(a\text{even}(x)\text{odd}(x), ax) = \text{permutation}(\text{even}(x)\text{odd}(x), x) = \text{permutation}(\text{odd}(x)\text{even}(x), x) = \text{true}$. Here, we used the induction hypothesis and the fact that xy is a permutation of yx (also straightforwardly proved with induction, if needed). $\square$

3.2 The human and monkeys theorem

Let's have a relation $<$, defined as follows: $x < y$ if $\text{parent}(x, y)$ or there exists z such that $x < z$ and $\text{parent}(z, y)$. We will assume that this relation is well-founded. Also, for all x , either $\text{human}(x)$ or $\text{monkey}(x)$.

Theorem 3.3. *If $x < y$, and $\text{human}(y)$, and $\text{monkey}(x)$, then there exist z_1, z_2 such that $\text{parent}(z_1, z_2)$, and $\text{human}(z_2)$, and $\text{monkey}(z_1)$.*

Proof. Define a sequence $w_1, w_2, \dots$ inductively like this: $w_1 = y$, w_{n+1} is such that $\text{parent}(w_{n+1}, w_n)$, and $x < w_{n+1}$. If $w_n = x$, then terminate the sequence. Obviously, $w_1 > w_2 > \dots$. Since $<$ is well-founded, the sequence has to be finite: $y = w_1 > w_2 > \dots > w_k = x$.

For the sake of contradiction, suppose, there is no such i , that $\text{parent}(w_{i+1}, w_i)$, and $\text{human}(w_i)$, and $\text{monkey}(w_{i+1})$. Then, by induction, all w_i are humans. This contradicts the fact that w_k is a monkey. Therefore, the desired z_1, z_2 exist among some w_{i+1}, w_i , as requested. $\square$

4 Homework 6: Soundness of FOL Resolution

In the following we use sets for both representing formulas in CNF and individual clauses. Hence, a formula in CNF is a set containing sets of literals.

We prove the soundness of the following FOL resolution rule:

$$\frac{C_1[l_1] \quad C_2[\neg l_2]}{C_1\theta[\perp] \vee C_2\theta[\perp]} \quad l_1\theta = l_2\theta$$

Lemma 4.1. *For any $C_1[l]$, $C_2[\neg l]$ first-order clauses and $\mathcal{I}$ interpretation, we get*

$$\mathcal{I} \models \{C_1[l], C_2[\neg l]\} \Rightarrow \mathcal{I} \models C_1[\perp] \vee C_2[\perp].$$

Proof. Let $\mathcal{I}$, $C_1[l]$, $C_2[\neg l]$ be arbitrary and $\mathcal{I} \models C_1[l]$ and $\mathcal{I} \models C_2[\neg l]$. We show $C_1[\perp] \vee C_2[\perp]$. Consider the two cases for $\llbracket l \rrbracket_{\mathcal{I}}$:

Case 1. $\llbracket l \rrbracket_{\mathcal{I}} = \text{true}$: Consider the clause $C'_2 = C_2 \setminus \{\neg l\}$, corresponding to the clause C_2 with $\neg l$ removed. Since $\llbracket \neg l \rrbracket_{\mathcal{I}} = \text{false}$, and $\mathcal{I} \models C_2[\neg l]$, $\mathcal{I} \models C'_2$. Moreover, since $\mathcal{I} \models C'_2$, $\mathcal{I} \models C'_2 \cup \{\perp\}$ and therefore $\mathcal{I} \models C_2[\perp]$. Finally, since $\mathcal{I} \models C_2[\perp]$, $\mathcal{I} \models C_1[\perp] \vee C_2[\perp]$.

Case 2. $\llbracket l \rrbracket_{\mathcal{I}} = \text{false}$: symmetric to Case 1. $\square$

Lemma 4.2. *Let $\mathcal{I}$, C be an arbitrary interpretation and clause, and σ a substitution.*

$$\mathcal{I} \models \forall x_1 \dots x_n, C \Rightarrow \mathcal{I} \models \forall x_1 \dots x_n, C\sigma$$

Proof. Let $\mathcal{I}, C, \sigma$ be arbitrary with $\mathcal{I} \models \forall x_1 \dots x_n C$, and assume towards a contradiction that $\mathcal{I} \not\models \forall x_1 \dots x_n, C\sigma$. By the definition of $\llbracket \forall x. \phi \rrbracket$, there must be a mapping $\kappa = [y_1 \rightarrow d_1, \dots, y_n \rightarrow d_n]$ from free variables in $C\sigma$ to $d_1 \dots d_n \in D_{\mathcal{I}}$ s.t. $\llbracket C\sigma \rrbracket_{\mathcal{I}[\kappa]} = \text{false}$.

We now construct a mapping $\kappa' = x_1 \rightarrow d'_1, \dots, x_n \rightarrow d'_n$ s.t. $\llbracket C \rrbracket_{\mathcal{I}[\kappa']} = \text{false}$. Consider the syntactic substitution $\sigma = x_1 \rightarrow t_1, \dots, x_n \rightarrow t_n$, we simply define $\kappa' = [x_1 \rightarrow t_1\kappa, \dots, x_n \rightarrow t_n\kappa]$. Hence, κ' is a map from free variables in C to elements in the domain $D_{\mathcal{I}}$ and is a valid mapping to close the formula C .

Moreover, $\llbracket C \rrbracket_{\mathcal{I}[\kappa']} = \llbracket C\sigma \rrbracket_{\mathcal{I}[\kappa]} = \text{false}$. Hence, $\mathcal{I} \not\models \forall x_1, \dots, x_n C$, a contradiction. $\square$

Theorem 4.3 (Soundness of FOL resolution).

For any clause set S , clauses $C_1[l_1] \in S$, $C_2[\neg l_2] \in S$, and substitution θ s.t. $l_1\theta = l_2\theta$: S is satisfiable iff $S \cup \{C_1\theta[\perp] \vee C_2\theta[\perp]\}$ is satisfiable.

Proof.

$\Rightarrow$ Let S , $C_1[l_1] \in S$, $C_2[\neg l_2] \in S$ s.t. $l_1\theta = l_2\theta$, be arbitrary and S satisfiable. By definition of satisfiable, let $\mathcal{I}$ s.t. $\mathcal{I} \models S$ and therefore $\mathcal{I} \models C_1$, and $\mathcal{I} \models C_2$. Note, that both clauses C_1 and C_2 are implicitly universally quantified, and since $\forall$ distributes over $\wedge$, we get $\mathcal{I} \models \forall x_1 \dots x_n, C_1$ and $\mathcal{I} \models \forall x_1 \dots x_n, C_2$ where $x_1 \dots x_n$ are the free variables in S .

By Lemma 4.2, $\mathcal{I} \models \forall x_1 \dots x_n, C_1\theta$ and $\mathcal{I} \models \forall x_1 \dots x_n, C_2\theta$. In the following we drop the implicitly quantified variables again. Let $l = l_1\theta = l_2\theta$ and since the order of substitution is irrelevant, $\mathcal{I} \models C_1[l]\theta$ and $\mathcal{I} \models C_2[l]\theta$. By Lemma 4.1, $\mathcal{I} \models C_1\theta[\perp] \vee C_2\theta[\perp]$, and therefore, by definition of satisfiable, $S \cup \{C_1\theta[\perp] \vee C_2\theta[\perp]\}$ is satisfiable.

$\Leftarrow$ Let S , $C_1[l_1] \in S$, $C_2[\neg l_2] \in S$ s.t. $l_1\theta = l_2\theta$, be arbitrary and $S \cup \{C_1\theta[\perp] \vee C_2\theta[\perp]\}$ satisfiable. By definition of satisfiable, let $\mathcal{I}$ s.t. $\mathcal{I} \models S \cup \{C_1\theta[\perp] \vee C_2\theta[\perp]\}$. By definition of $\wedge$, $\mathcal{I} \models S$ and therefore S is satisfiable. $\square$

5 Homework 9

5.1 Formulate QBF as a first-order theory

We define the theory T_{QBF} over the signature $\langle \{t^0, f^0\}, \{v^1\} \rangle$ with constant function symbols t and f and a unary predicate v .

5.1.1 Define a first-order theory T_{QBF}

We axiomatize the theory:

1. $\forall x, x = t \vee x = f$
2. $t \neq f$
3. $v(t)$
4. $\neg v(f)$

The first two axioms restrict the models to exactly those models with a domain of cardinality of two. The third and fourth axiom ensure that the constant t is interpreted as true and f as false.

5.1.2 Define a function f from closed QBF formulas to T_{QBF} formulas s.t. a formula ϕ is true iff $f(\phi)$ is satisfiable.

Let $\mathcal{P}$ be the set of propositions. We define a function $f : \mathcal{P} \rightarrow \mathcal{A}$ from propositions to atomic formulas. We consider formulas as defined in the lecture and add existential quantification and $\top$ as they are used in QBF.

- $f(\top) \mapsto \top$
- $f(\perp) \mapsto \perp$
- $f(p) \mapsto v(p)$

We then get a function $\mathcal{F} : \psi_{QBF} \rightarrow \psi_{T_{QBF}}$ by mapping f over the connectives of FOL.

Lemma 5.1. *For any closed QBF formula ϕ , ϕ is true $\iff \mathcal{F}(\phi)$ is satisfiable.*

Proof. We will consider the natural structure $\mathcal{I} := \langle \{0, 1\}, \{t, f\}, \{v\} \rangle$ with $t_{\mathcal{I}} = 1$, $f_{\mathcal{I}} = 0$ and $v = \{1\}$. We omit writing the subscript $\mathcal{I}$ in the following. Clearly, the structure is a T_{QBF} -Model. We prove that ϕ is true iff $\mathcal{I} \models \mathcal{F}(\phi)$ by induction on the structure of ϕ .

- $\phi := \perp$. For both directions we get an immediate contradiction as $\mathcal{I} \not\models \perp$ and $\perp$ is not true.
- $\phi := \top$, trivially $\mathcal{I} \models \mathcal{F}(\top) = \top$ and $\top$ is true, hence $\top$ is true iff $\mathcal{I} \models \mathcal{F}(\top)$.
- $\phi := p$, (for completeness, ϕ is closed)
- $\phi := \phi_1 \Rightarrow \phi_2$. Assume ϕ is true, then either ϕ_1 is false or both ϕ_1 and ϕ_2 are true. By induction, in the first case we get $\mathcal{I} \not\models \mathcal{F}(\phi_1)$, and therefore $\mathcal{I} \models \mathcal{F}(\phi)$ and in the second case $\mathcal{I} \models \mathcal{F}(\phi_1)$ and $\mathcal{I} \models \mathcal{F}(\phi_2)$, hence $\mathcal{I} \models \mathcal{F}(\phi)$. For the other direction, assume $\mathcal{I} \models \mathcal{F}(\phi)$. Then either $\mathcal{I} \not\models \mathcal{F}(\phi_1)$ or both $\mathcal{I} \models \mathcal{F}(\phi_1)$ and $\mathcal{I} \models \mathcal{F}(\phi_2)$. By induction, in the first case ϕ_1 is false, hence ϕ true and in the second case both ϕ_1 and ϕ_2 are true, hence ϕ is true.
- $\phi := \forall \phi_1$. By (IH) we know that $\phi_1[\top]$ is true iff $\mathcal{I} \models \mathcal{F}(\phi_1[\top])$ and $\phi_1[\perp]$ is true iff $\mathcal{I} \models \mathcal{F}(\phi_1[\perp])$. We continue to show that $\mathcal{I} \models \mathcal{F}(\phi_1[\top])$ iff $\mathcal{I}_{x \leftarrow 1} \models \mathcal{F}(\phi_1)$ and $\mathcal{I} \models \mathcal{F}(\phi_1[\perp])$ iff $\mathcal{I}_{x \leftarrow 0} \models \mathcal{F}(\phi_1)$, hence by definition of $\forall$, $\mathcal{I} \models \mathcal{F}(\phi)$

case $\top$: Since $\mathcal{I} \models \top \iff v(t)$, we know $\mathcal{I} \models \mathcal{F}(\phi_1[v(t)])$ iff $\mathcal{I} \models \mathcal{F}(\phi_1[\top])$ (*Replacement theorem* would require another (trivial) proof by induction on formula). Moreover, $\mathcal{I} \models \mathcal{F}(\phi_1[v(t)]) \iff \mathcal{I}_{x \leftarrow 1} \models \mathcal{F}(\phi_1[v(x)])$ and $\phi_1[v(x)] = \phi_1$. By transitivity of $\iff$, $\phi_1[\top]$ is true iff $\mathcal{I}_{x \leftarrow 1} \models \mathcal{F}(\phi_1)$.

case $\perp$: Symmetric to **case $\top$** .

- $\phi := \exists \phi_1$. By definition, either $\phi_1[\perp]$ or $\phi_1[\top]$ is true. We proceed by cases, by (IH) we know that in the first case $\phi_1[\top]$ is true iff $\mathcal{I} \models \mathcal{F}(\phi_1[\top])$ and in the other case $\phi_1[\perp]$ is true iff $\mathcal{I} \models \mathcal{F}(\phi_1[\perp])$. For each case, the proof is identical to the cases $\perp$ and $\top$ above.

To prove the final statement, assume ϕ is true, then by the proof above $\mathcal{F}(\phi)$ is satisfiable as $\mathcal{I} \models \mathcal{F}(\phi)$. For the other direction, assume $\mathcal{F}(\phi)$ is satisfiable, then there exists an $\mathcal{I}'$ s.t. $\mathcal{I}' \models \mathcal{F}(\phi)$. By the axioms, $\mathcal{I}'$ is isomorphic to $\mathcal{I}$, since $\mathcal{I}'$ is a T_{QBF} -Model and therefore by axioms (1) and (2) has a 2 element domain, with the element $x = \llbracket t \rrbracket'_{\mathcal{I}'}$ being the only element s.t. $x \in \llbracket v \rrbracket'_{\mathcal{I}'}$. Therefore for any ψ , $\mathcal{I}' \models \psi$ iff $\mathcal{I} \models \psi$. Hence, also $\mathcal{I} \models \mathcal{F}(\phi)$ and therefore ϕ is true. $\square$

5.2 Argue that Presburger arithmetic does not allow QE.

To argue informally that Presburger arithmetic does not allow quantifier elimination in general, we need to demonstrate that there exists a formula in Presburger arithmetic where quantifier elimination is not possible. We will show this by identifying an example formula that cannot be simplified to an equivalent formula without quantifiers.

Consider the formula $\phi := \exists k, k + k = n$, the key point is that:

- The formula $\exists k(2k = n)$ expresses the idea that n is even.
- In Presburger arithmetic, there is no built-in way to express **divisibility by a constant** like 2 without quantifying over possible values of k .

Any attempt to eliminate the quantifier would require some form of induction or recursion, which is not possible in the setting of Presburger arithmetic (since it only allows addition and equality, and not multiplication or recursion). Thus, the formula $\exists k(2k = n)$ **cannot** be expressed without quantifiers in Presburger arithmetic.